✅ NYAYA SETU (MVP) — Detailed Step-by-Step Build Plan (Tasks + Subtasks + Dependencies)

This plan is based on the final PRD and includes the **4 UI refinement goals**:

1. Navbar + Dark Mode polish
2. Premium homepage + cards
3. Results page better layout
4. Loading + error screens improvement

✅ Every task has dependencies
✅ No orphan tasks
✅ Each step is small + shippable
✅ Follows best practices (incremental delivery, low risk jumps)

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0) Project Initialization & Standards

✅ Task 0.1 — Repository + baseline tooling

Depends on: None

- Create git repo and push initial commit
- Add .gitignore, .editorconfig, README.md
- Configure TypeScript strict mode (or confirm it is enabled)
- Add ESLint + Prettier config
- Add scripts in package.json:
  - dev
  - build
  - lint
  - format

✅ Task 0.2 — Define shared types + constants (PRD contract)

Depends on: 0.1

- Create shared/constants.ts:
  - Allowed file types: PDF/JPG/PNG
  - Max upload size: 10MB
  - Supported languages (8)
  - Max stored docs: 3
  - Category buckets (5 + Other/Unclear)
- Create shared/types.ts:
  - Language
  - UrgencyLevel (Low/Medium/High)
  - ExtractedFields (sender/receiver/date/deadline/subject/demands)
  - DecodedResult (docType, category, urgency, summary, keyPoints, actionPlan)
  - StoredDoc (id, title, createdAt, preview, decodedResult)

✅ Task 0.3 — Route plan + flow rules (forward-only)

Depends on: 0.2

- Define official routes:
  - /login
  - /home
  - /upload
  - /preview
  - /language
  - /confirm
  - /loading
  - /result/:id
- Write rules for accessing pages:
  - No auth → redirect to /login
  - No upload session → block preview/language/confirm/loading
  - No decode result → block result view

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1) UI Foundation (Premium Baseline)

✅ Task 1.1 — Tailwind base design system + layout wrapper

Depends on: 0.1

- Ensure TailwindCSS is installed and working

- Set darkMode: "class" in Tailwind config
- Create common design helpers:
    - Card styling utility class
    - Badge styling utility class
    - PrimaryButton / SecondaryButton styling rules
- Build AppShell component:
    - max width container
    - consistent padding
    - minimum height full screen
    - background support light/dark

✅ Task 1.2 — Add “Page template” component for consistent screens

Depends on: 1.1

- Create Page.tsx wrapper component:
    - title slot
    - optional description slot
    - children container
- Apply Page wrapper to placeholder screens for visual consistency

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2) Feature Set 1/4 — Navbar + Dark Mode Polish ✅

✅ Task 2.1 — Build Navbar component (minimal, premium)

Depends on: 1.2

- Create Navbar.tsx
- Navbar must display:
    - App name: NYAYA SETU
    - Tagline: Legal clarity for everyone.
    - Right side area for actions (theme toggle, logout)
- Keep minimal height, clean spacing, and consistent borders

✅ Task 2.2 — Implement Dark Mode toggle (manual)

Depends on: 2.1

- Create ThemeToggle.tsx
- Store preference in localStorage
- Toggle dark mode by adding/removing dark class on <html>
- Ensure toggle exists only on /home (homepage-only rule)

✅ Task 2.3 — Verify dark mode compatibility across core UI elements

Depends on: 2.2

- Ensure readable contrast for:
  - navbar
  - cards
  - badges
  - buttons
- Fix any illegible text in dark mode

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3) Authentication (Mandatory OTP)

✅ Task 3.1 — Setup OTP provider config (Firebase recommended)

Depends on: 0.1

- Create Firebase project
- Enable Phone OTP authentication
- Add .env values (keys/config)
- Implement Firebase init utility (firebase.ts)
- Add error-safe config loading

✅ Task 3.2 — Build Login screen (Phone → OTP → Verify)

Depends on: 3.1, 1.2

- Create /login UI:
  - phone input
  - send OTP button
  - OTP input
  - verify button

- Handle auth errors cleanly (toast/message)
- Redirect to /home on success

✅ Task 3.3 — Session + logout + auth guard

Depends on: 3.2, 0.3

- Create useAuth() hook:
  - current user
  - loading state
- Protect routes:
  - unauthenticated → /login
- Add logout action in Navbar (visible when logged in)
- Ensure session persists ~30 days

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4) Local Persistence (IndexedDB) + “Latest 3 Only”

✅ Task 4.1 — Implement IndexedDB database layer (Dexie)

Depends on: 0.2

- Install Dexie
- Create db.ts with tables:
  - documents
- Implement DB methods:
  - saveDocument(doc)
  - getDocumentById(id)
  - listRecentDocuments(limit=3)
  - deleteDocumentById(id)

✅ Task 4.2 — Enforce “keep only last 3 docs” rule

Depends on: 4.1

- Implement enforceMaxDocuments(3)
- On every new save:
  - if count > 3 → delete oldest
- Add dev logs to confirm deletion happens correctly

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5) Feature Set 2/4 — Homepage Premium + Cards ✅

✅ Task 5.1 — Build Homepage screen layout

Depends on: 2.3, 3.3

- Create /home screen with:
  - Navbar (with theme toggle)
  - Primary CTA: **Scan & Decode**
  - Recent documents section placeholder
- Hook CTA to /upload

✅ Task 5.2 — Premium Recent Docs cards (max 3)

Depends on: 4.2, 5.1

- Fetch recent docs from IndexedDB (limit 3)
- Render card per document:
  - title
  - created date
  - detected type badge (if available)
- Clicking card opens /result/:id

✅ Task 5.3 — Homepage empty state UX

Depends on: 5.2

- If no docs:
  - show helpful message (“Upload a document to begin”)
  - show CTA prominently

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6) Upload Screen (PDF + Images)

✅ Task 6.1 — Build Upload screen UI + validation

Depends on: 5.1, 0.2

- Create /upload screen with:
  - upload PDF button
  - upload images button (multi-select)
- Validate:
  - allowed types only
  - total size <= 10MB
- Show error messages on invalid input

✅ Task 6.2 — Create decode session state store

Depends on: 6.1

- Implement useDecodeSession() store:
  - selected files
  - file type (pdf/images)
  - preview edits
  - selected language
  - decode status
- Persist only in memory (session-level)
- After upload validated → go to /preview

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7) Preview + Crop/Rotate

✅ Task 7.1 — Build Preview screen

Depends on: 6.2

- Create /preview screen:
  - display preview of uploaded document
  - show crop and rotate controls
- Add Continue button → /language

✅ Task 7.2 — Save crop/rotate edits into decode session

Depends on: 7.1

- Apply crop/rotate transformations to stored images

- Store final edited version in session state
- Ensure edited version is used for decoding

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8) Language Selection (Every Decode)

✅ Task 8.1 — Build Language selection screen

Depends on: 7.2, 0.2

- Create /language screen:
  - dropdown of 8 languages
  - selection required
- Save selected language into decode session state
- Continue → /confirm

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9) Confirm Upload + Warning

✅ Task 9.1 — Build Confirm screen + keep-last-3 warning

Depends on: 8.1

- Create /confirm screen showing:
  - file name(s)
  - selected language
- Show warning message:
  - “Only your latest 3 documents are kept.”
- Proceed button → /loading

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10) Feature Set 4/4 — Loading + Error Screens Improvement ✅

✅ Task 10.1 — Build Loading screen polish

Depends on: 9.1

- Create /loading screen:

    - premium centered spinner
    - “Decoding…” text
- Ensure UI doesn’t freeze during processing

✅ Task 10.2 — Build Decode failure UI + scan tips

Depends on: 10.1

- If decode fails:
    - show “Could not decode reliably”
    - show “Re-upload” button → /upload
    - show scan tips list (lighting, blur, crop, shadows)

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11) OCR + Decode API Pipeline

✅ Task 11.1 — Implement OCR module (OCR everything)

Depends on: 9.1

- Create OCR wrapper:
    - PDF → OCR extraction (even if selectable text exists)
    - Images → OCR extraction
- Return extracted text as a single string
- Add timeouts and max text length cap

✅ Task 11.2 — Implement LLM decoder (strict output schema)

Depends on: 11.1, 0.2

- Create decoder prompt to output:
    - detected doc type
    - category tag
    - urgency label
    - extracted details fields
    - Summary (5–7 lines)
    - Key Points (6–10 bullets)
    - Action Plan = “Consult a lawyer.”
- Validate output schema (reject malformed output)

✅ Task 11.3 — Build `/api/decode` endpoint

Depends on: 11.2

- Accept payload:
  - uploaded file(s)
  - selected language
- Execute:
  - OCR → text
  - LLM decode → structured output
- Return JSON result to frontend

✅ Task 11.4 — Connect Loading screen to decode endpoint

Depends on: 11.3, 10.1, 10.2

- On /loading mount:
  - call /api/decode
  - success → store decoded output
  - failure → show failure UI + scan tips

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12) Save Decoded Document Locally (and enforce latest 3)

✅ Task 12.1 — Save decoded doc into IndexedDB

Depends on: 11.4, 4.2

- Create StoredDoc object:
  - id
  - createdAt
  - default title
  - preview reference (first page)
  - decoded result

    - extracted raw OCR text
- Save to IndexedDB
- Enforce keep-only-last-3 rule
- Redirect to /result/:id

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13) Feature Set 3/4 — Results Page Better Layout ✅

✅ Task 13.1 — Results screen structure + warning banner

Depends on: 12.1

- Create /result/:id
- Load stored document from IndexedDB
- Show top banner:
    - “This may be inaccurate.”

✅ Task 13.2 — Add premium “Details” card + urgency label

Depends on: 13.1

- Show extracted fields at top:
    - sender, receiver, date, deadline, subject, demands
- Show “Reply by: ”
- Show urgency badge (Low/Medium/High)
- Show detected doc type + category tag badges

✅ Task 13.3 — Tabs UI: Summary / Key Points / Action Plan

Depends on: 13.2

- Add tabs:
    - Summary (5–7 lines)
    - Key Points (6–10 bullets)
    - Action Plan: “Consult a lawyer.”

✅ Task 13.4 — Document preview (first page) + modal

Depends on: 13.2

- Render preview thumbnail
- On click open modal viewer
- Show only first page

✅ Task 13.5 — Collapsible “Original Extracted Text”

Depends on: 13.3

- Add collapsible section below tabs
- Render OCR text read-only
- Do not add copy buttons

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14) Rename + Delete + Undo (5 seconds)

✅ Task 14.1 — Rename document

Depends on: 13.1

- Add rename button in results page
- Modal input for title
- Persist title change to IndexedDB
- Ensure homepage recent cards reflect new title

✅ Task 14.2 — Delete document with confirm + undo toast

Depends on: 14.1

- Add delete button
- Confirmation modal required
- On delete:
    - remove from IndexedDB
    - show Undo toast (5 seconds)
- If Undo clicked:
    - restore document
- After final delete:

- redirect to /home

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15) Export PDF (Single report)

✅ Task 15.1 — Implement Export PDF report

Depends on: 13.4, 13.3

- Add export button on results page
- Generate single PDF including:
  - document preview (first page)
  - extracted details fields
  - Summary + Key Points + Action Plan
- Ensure:
  - no watermark
  - normal PDF download
  - no disclaimer inside PDF

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16) Rating + Feedback (KPI)

✅ Task 16.1 — Add rating widget + optional text feedback

Depends on: 13.3

- Place rating section on results screen
- Allow 1–5 selection
- Optional feedback text
- Save feedback locally (IndexedDB) or POST to endpoint

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17) QA + Demo Hardening

✅ Task 17.1 — End-to-end QA checklist run

Depends on: 16.1, 15.1, 14.2, 10.2

- Login works (OTP)
- Upload PDF works

- Upload images works
- Preview crop/rotate works
- Language selection works (8 languages)
- Decode flow works (up to 60 seconds)
- Results show proper tabs, details, preview, extracted text
- Save-only-last-3 rule works (4th deletes oldest)
- Rename works
- Delete + Undo works
- Export PDF works
- Failure screen works + scan tips show properly
- Dark mode works and looks premium

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18) Deployment

✅ Task 18.1 — Deploy + verify production stability

Depends on: 17.1

- Add production env vars (auth + OCR + LLM)
- Deploy to hosting (Replit publish / Vercel)
- Run production smoke test:
  - login
  - decode
  - export PDF
  - open recent docs

✅ Dependency Graph (Simple Summary)

- Foundation: **0.1** → **0.2** → **0.3**
- UI Base: **1.1** → **1.2**
- Navbar/Dark Mode: **2.1** → **2.2** → **2.3**
- Auth: **3.1** → **3.2** → **3.3**
- Storage: **4.1** → **4.2**
- Homepage: **5.1** → **5.2** → **5.3**
- Upload Flow: **6.1** → **6.2** → **7.1** → **7.2** → **8.1** → **9.1**
- Loading/Error: **10.1** → **10.2**

- Decode Pipeline: **11.1** → **11.2** → **11.3** → **11.4**
- Save Decoded: **12.1**
- Results Polish: **13.1** → **13.2** → **13.3** → **13.4** → **13.5**
- Actions: **14.1** → **14.2**
- Export: **15.1**
- KPI: **16.1**
- QA + Deploy: **17.1** → **18.1**
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